Leading University

Department of Computer Science and Engineering

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Project report on User Management System

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1. Project Title

User Management System Using HTML, CSS, PHP, MySQL, Bootstrap, JavaScript and PHPMailer

2. Introduction

The User Management System is a web-based application developed using PHP and MySQL. The system manages user registration, login authentication, profile management, and user activity tracking.

3. Objectives

The objectives of this project are:

1. To create a secure user registration and login system
2. To perform CRUD operations
3. To implement session-based authentication
4. To use PHPMailer for email sending
5. To validate user input using JavaScript
6. To maintain login history
7. To develop a responsive user-friendly interface

4. Technologies Used

The Technologies in this project are:

1. HTML for Structure of web pages
2. CSS for Styling
3. Bootstrap for Responsive design
4. JavaScript for Client-side validation and Dark Mode
5. PHP for Backend development
6. MySQL for Database management
7. PHPMailer for Sending confirmation emails

5. System Features

5.1 User Registration

Users can create an account by providing:

1. Name
2. Email
3. Password
4. Phone Number

The system checks for duplicate emails before registration.

5.2 Email Confirmation

After successful registration, a confirmation email is sent to the user using PHPMailer and Gmail SMTP.

5.3 User Login

1. Registered users can login using their email and password.
2. The system verifies user credentials from the database.

5.4 Session Management

1. PHP sessions are used to keep users logged in securely.
2. Unauthorized users cannot access the dashboard page directly.

5.5 Dashboard

The dashboard displays all registered users in a table format.

Features are:

1. View User Information
2. Edit User Information
3. Delete User

5.6 CRUD Operations

The project supports complete CRUD functionality:

1. Create for Register new user
2. Read for Display user data
3. Update for Edit user information
4. Delete for Remove user

5.7 Frontend Validation

JavaScript validation is used to:

1. Check password length
2. Validate phone number length
3. Prevent empty form submission

5.8 Dark/Light Mode

The project includes a Dark/Light mode feature implemented using JavaScript and Bootstrap classes.

5.9 Login History

Every successful login is stored in the login_history table with timestamp information. This helps track user activity.

6. Database Design

Database Name is user_management.

Table: users

Field Name	Data Type
id	INT (Primary Key)
name	VARCHAR(100)
email	VARCHAR(100)
password	VARCHAR(100)
phone	VARCHAR(20)

Table: login_history

Field Name	Data Type
id	INT (Primary Key)
email	VARCHAR(100)
login_time	TIMESTAMP

7. System Workflow

User Registration
↓
Data Stored in Database
↓
Confirmation Email Sent
↓
User Login
↓
Session Created
↓
Dashboard Access
↓
Edit/Delete Operations
↓
Login History Stored
↓
Logout

8. Implementation Details

Frontend

The frontend was developed using:

1. HTML
2. CSS
3. Bootstrap
4. JavaScript

Bootstrap was used to create responsive forms, tables, and buttons.

JavaScript was used for form validation and Dark/Light mode functionality.

Backend

PHP was used for backend development.

Main backend functionalities include:

1. Database connection
2. Form handling
3. Session management
4. CRUD operations
5. Authentication
6. Email integration

Database

MySQL database was used to store:

1. User information
2. Login history

9. Usage Instructions

Step 1

Start Apache and MySQL from XAMPP.

Step 2

Create the database: user_management

Import the required tables.

Step 3

Place the project folder inside: htdocs

Step 4

Open the browser and run:

http://localhost/user_management

Step 5

Register a new account.

Step 6

Login using registered credentials.

Step 7

Access the dashboard to manage users.

10. Challenges Faced

During development, some challenges were faced:

1. SMTP configuration for PHPMailer
2. Session handling
3. JavaScript validation
4. Database connection setup

These issues were solved through testing and debugging.

11. Conclusion

The User Management System successfully demonstrates user authentication, CRUD operations, session handling, email integration, and responsive web design using PHP and MySQL.

The project provides a practical implementation of web development concepts and improves understanding of frontend and backend integration.

12. Future Improvements

Future improvements may include:

1. Password hashing
2. Email verification links
3. Forgot password system
4. Profile picture upload
5. Admin panel
6. Advanced security features